

General Purpose, Information, Intelligence: AI Governance and the Anatomy of a Race

Every field of engineering eventually discovers that the hard part is not the machine. Nuclear physics was solved in a laboratory; the problem of what to do about it has occupied diplomats for eighty years and is not finished. So far in this course we have treated the world around the model as a black box. We have said that models get built, get deployed, and that somebody, somewhere, decides how. This lecture opens that box — the norms, laws, institutions, and rivalries that shape what gets built and who benefits — and asks why it is so much harder to close than the model itself.

The essay proceeds in three movements. First, the stakes: what AI governance is, and the four clusters of risk that motivate the field. Second, a set of three theoretical lenses — AI as a general purpose technology, as an information technology, and as an intelligence technology — each of which supplies a reference class of historical analogies and a mechanism for predicting what AI will do. Third, the anatomy of governance itself: institutions, externalities, and the one structural condition, anarchy among great powers, that makes the most important problems the least tractable. If you can explain why a rational leader who believes the prize is existential should be willing to accept a one-in-three chance of nuclear war, you understand the strategic problem of AI governance.

The material draws throughout on Allan Dafoe’s introductory chapter to the *Oxford Handbook on AI Governance* (2022), which should be read in full; what follows is the lecture, not the chapter.

I. What Governance Is, and Why the Stakes Are High

Three terms get blurred and should not be. **AI policy** is what governments do. **AI ethics** is what practitioners ought to do. **AI governance** is the widest of the three: descriptively, every policy, norm, law, and institution that shapes how AI is built and deployed; normatively, the aspiration that those structures produce good decisions — effective, safe, inclusive, legitimate, adaptive. Governance is not a synonym for regulation. A publishing norm at a lab, a procurement standard, a hiring practice, an export control, and a treaty are all governance. The field, in one formulation, studies how humanity navigates the transition to advanced AI.

Why care so much? Two time horizons. In the coming **years**, AI will be deployed across transportation, health, news, education, science, policing, and the military, and because it is a general purpose technology the effects will be broad, deep, and hard to foresee. In the coming **decades**, the effects could be of a different order — a transformation of welfare, wealth, and power larger than the industrial revolution. When the chapter was written, surveyed ML researchers put roughly even odds on broadly human-level AI by 2060, and the most careful compute-based forecast landed in the same place. You do not need to believe the forecasts to see the point: if there is a serious chance the technology is that consequential, the institutions around it deserve as much attention as the technology.

Four risk clusters

To make “high stakes” concrete, the chapter names four clusters of extreme risk. Hold them in mind; each of the lenses below will explain where one or more of them comes from.

Inequality, turbulence, authoritarianism. A falling labor share of value and winner-take-

most markets could erode the rough equality that democracies rest on. AI-filtered communication could deepen polarization and manipulation. And the tools of a totalitarian state — ubiquitous surveillance, lie detection, autonomous weapons — get much better.

Great-power war. AI could make crises faster than humans can manage, a “flash war” by analogy to a flash crash, and could create first-strike advantages and power shifts that make war more likely.

Control, alignment, and political order. The **control problem** is the problem of human intention actually governing what an advanced AI does; the **alignment problem**, a component of it, is building agents whose goals are the goals of their principal. If out-of-control AI sounds implausible, start instead from an older, unsolved problem: controlling corporations, militaries, and parties. Our solutions there are patchwork and periodically fail. The AI version is analogous, and as AI empowers those same social entities the two problems compound.

Value erosion from competition. A high-stakes race pushes every participant to cut corners. Generalize: wherever something of value trades off against competitive advantage, competition erodes the value. Safety is the obvious casualty, but so are privacy, sustainability, decentralization, and equality. In the long run, ungoverned competition selects for whatever is most adaptive rather than what is good.

A holistic sensibility

A question from the audience: is this the “existential risk” school or the “AI ethics” school? The chapter’s answer is that the conflict is mostly misplaced. The scholarship needed for near-term harms and for extreme ones overlaps heavily, and the field should aim at **scalable governance**: solutions to today’s pressing problems that will still matter for tomorrow’s extreme ones. This mirrors what happened in AI safety, which began with thought experiments about superintelligence and now runs empirical research programs on existing systems. The missing puzzle pieces for any one problem are scattered across many disciplines. Be inclusive about motivations; be ambitious about scope.

II. Three Lenses

How should we think about what AI will do? Any framework must balance abstraction (so the vocabulary generalizes) against structure (so it makes predictions), parsimony against completeness, and it should rest on a microfoundation — a mechanism, not just a list of impacts. Lists are what most of the literature offers: AI delegates decisions to machines, accelerates growth, displaces labor, shifts the offense-defense balance, enables surveillance. All true; none explains itself.

The chapter offers three lenses instead, each of which supplies a mechanism and a reference class of prior technologies to learn from. Each later lens is, loosely, a special case of the one before it. Treat every claim below as a hypothesis, stated concisely, to be tested.

Lens 1: AI as a general purpose technology

A **general purpose technology** (GPT — the older meaning of the acronym) is one that (1) provides a valuable input to many processes and (2) enables important complementary innovations. Printing, steam, rail, electricity, motor vehicles, aviation, computers. Almost all of them are about energy, transportation, or information. GPTs are what economists reach for to explain long-run

growth.

AI is a GPT, and plausibly the quintessential one: a fundamental input to nearly everything, endlessly complementary with other processes. Kevin Kelly’s forecast for the next ten thousand startups — take X and add AI — is the whole lens in a sentence. Two features make AI more transformative than most GPTs: it is still early and capable of ongoing improvement, with a ceiling that likely exceeds human level; and the pace is faster than for any prior GPT.

What does the reference class predict? Five things.

1. **Growth**, often radical, through two channels: **efficiency** (cheaper inputs to existing processes) and **enabling** (new processes altogether). In principle this is Pareto-improving — if losers are compensated, everyone can gain.
2. **Disruption** of existing processes and therefore of the social relations built on them: distributional consequences, displacement costs, and losers whose losses are hard to identify and therefore hard to insure. Early GPTs look harmless — a cumbersome press, a slow prototype railway — and only become revolutionary after generations of improvement.
3. **Mobilization of anticipated losers.** Workers, firms, and asset holders who fear displacement resist, and the resistance ranges from minor regulation to revolution. Internationally, the anticipated rise and fall of countries can precipitate war.
4. **Strategic importance.** Some GPTs are essential to the military-industrial base; possessing the mature version is close to a necessary condition of great-power status. They become sites of investment and rivalry.
5. **Dual use.** Peaceful and military applications are hard to separate, which is exactly what makes arms control difficult.

None of this is new to AI. That is the point of a reference class: two centuries of experience with other GPTs is available to learn from.

A caveat the chapter flags and then returns to. The net impact of those two centuries has favored labor and liberal institutions. But that arguably held because labor and liberal institutions were economic and military **complements** to the new technology — an educated, free citizenry was more productive. Nothing guarantees that continues.

Lens 2: AI as an information technology

An **information technology** improves the production, compression, transmission, reproduction, enhancement, storage, control, or use of information. AI improves every item on that list and, because they complement each other, amplifies every other information technology. It compresses a video stream of a public square into a record of who was where and when; it makes data searchable across modalities; it opens trillions of dollars of new uses.

Some information technologies were GPTs and inherit the properties above — writing for civilization, printing for the nation-state, telegraph and radio for modern war and commerce. But information technologies have extra properties of their own, and three deserve names.

Increasing returns and distribution. Information has low marginal cost relative to fixed cost — in ML terms, training a model costs orders of magnitude more than serving one more user —

and networks are worth more the more people are on them. Both push toward concentrated market structures and winner-take-most labor markets, and therefore toward **income** inequality. But there is a countervailing pull toward **consumption** equality, because information wants to be free: it is non-rival, hard to exclude, and cheap to steal. The efficient price is marginal cost, near zero, and public-interest services, IP limits, and competition all push prices there. Consider a billionaire: the books, films, games, maps, and social feeds they use are almost all available to the median earner.

Coordination and identity. Information technology can broaden identities (national identity out of local ones; cosmopolitanism out of global media) or narrow them by letting spatially dispersed people with a common interest find each other — global ideologies, religions, movements, gig economies. Whether AI strengthens or undermines existing communities is not determined by the lens; the lens tells you where to look.

Power. Information shifts bargaining power (information rents), decides adversarial contests (the US intelligence budget is a tenth of its military budget; coups turn on who learns what, when), and enables both democratic oversight and totalitarian suppression. Centralization tracks the authority's ability to monitor its agents: the telegraph curtailed the autonomy of ambassadors and ship captains; autonomous weapons let commanders bypass officers who might refuse. And information moves faster than everything else, which is why crises accelerate — the flash crash is the model for the flash war.

The net effect on politics is not knowable in advance. Whether AI strengthens the state, weakens it, or transforms it is open. What is not open is that information is a critical political resource and AI multiplies its value.

Lens 3: AI as an intelligence technology

The third lens is the least developed in the literature and, the chapter argues, the most important. An **intelligence technology** is an innovation in some entity's ability to solve cognitive tasks. The reference class is much larger than machines. Place these on a spectrum.

At one end are **tools**: an abacus, a dictionary, a notepad. Narrow, non-autonomous; they need a user to have any effect. At the other end are **systems** and **agents**: general and autonomous, with agents being those that behave like coherent goal-seekers. Systems include the price mechanism, language, bureaucracy, peer review, the law. Agents include the grand vizier, the general staff, the corporation. Markets and bureaucracies are artificial intelligences in this sense, and they have alignment and control problems too.

Intelligence technologies share the properties of the earlier lenses — critical for survival, transformative of the largest political entities (the tribe, the Neolithic state, the modern state each arose partly from one) — and add a distinctive question: do they **complement** or **substitute** for human cognitive labor? Calculators substituted for human calculators and complemented mathematicians. Bureaucracies substitute for a minister and complement a good decision-maker. Whether future AI complements or substitutes is the variable that drives labor share, inequality, and growth rates, because capital that can think can grow itself.

Bias, alignment, control

Now the payoff of the third lens. Every intelligence technology, however simple, poses one of three problems in ascending order of severity.

Bias. Even a tool bends decisions toward what fits it: to a person with a hammer, everything looks like a nail. Early states favored legible social arrangements; policymakers over-weight GDP; social media optimized for engagement. Because most decisions have political consequences, so do the tools that shape them — AI in hiring and sentencing, but also email and Twitter in how people deliberate.

Misalignment. A system or agent may not share the principal’s interests, so increasing its power systematically produces outcomes the principal did not want. Markets misallocate when externalities are large. Enron’s accounting, in collusion with its auditor, is agent misalignment. Civilian control of the military periodically fails — Kennedy and the Joint Chiefs during the missile crisis.

Loss of control. The principal-agent problem in its hardest form, where the agent’s advantage is not merely informational but a potentially vast gap in intelligence. The principal may not know what questions to ask or possess the concepts to frame the problem.

The social sciences have been working on this for centuries: oversight, transparency, whistleblowers, representation, institutional design. AI alignment is currently done almost entirely by AI researchers. Given the consilience, it should not be.

III. Governance and Anarchy

Now the black box. Governance means shaping behavior toward social goals through institutions, where “institutions” covers the full spectrum from norms and rituals to regulators and legislatures. The one conceptual tool you need from economics is the **externality**: a by-product of one actor’s behavior on others, positive or negative. The central insight is that institutions raise welfare by discouraging negative externalities (pollution) and encouraging positive ones (innovation). Institutions exist to internalize externalities.

Institutional fit

That gives a method. Confronted with a governance problem, ask what externalities need internalizing, over what political space they span, and whether any existing or imaginable institution has the needed:

- spatial remit;
- issue-area remit;
- political remit — does it legitimately represent the relevant stakeholders?
- technical competence;
- institutional competence;
- influence — can it actually move material incentives?

Worked example: self-driving cars. Interests: citizen safety, traveler mobility, producer profits. Primary externality: the effect of driving algorithms on other road users. Existing traffic-safety agencies have a default presumption of fit — and then you ask what is actually different. Safety must be evaluated from fleet statistics and simulators rather than driving tests. Liability must be allocated between producer and user. Insurance pricing depends on who was driving and what sensor

data was shared. There are new risks (hacking) and new **scales** of risk (a whole fleet hacked at once). There are coordination opportunities — cooperative algorithms, standards, smart infrastructure — and trade tensions around harmonization. And some issues will absorb public attention out of all proportion to their importance: the trolley problem is philosophically gripping and nearly irrelevant to the work of improving welfare. Having done the analysis, you can diagnose the institution you need.

The rule that follows: the further a new governance area sits from an existing, legitimate, competent institution, the harder adaptation will be. Some public goods fail purely on this distance. Every out-of-copyright book, free at every library, is not provided — not for want of anyone willing to scan them, but because Congress cannot update copyright law.

Deeply politicized issues

Adaptation is hardest of all when an issue falls inside an unresolved social conflict, because then there is no consensus on goals, no legitimate institution to work with, and the issue itself becomes a battleground for the larger fight. Call these **deeply politicized** governance issues. Two arenas.

Domestic conflicts. Social media moderation sits squarely inside the American left-right cleavage: both sides fear censorship, bias, fake news, and foreign influence, and disagree about what a legitimate moderator would even look like. Algorithmic fairness is the sharper case. ProPublica reported that a widely used recidivism classifier had a lower false-positive rate for white defendants than for Black defendants. Later work showed that if any demographic difference in false positives, false negatives, or calibration counts as bias, then — for an imperfect classifier facing different base rates — bias is mathematically unavoidable. Under that definition every classifier is biased, so the definition must be refined. The lesson is not that the problem is fake; it is that a new capability forced us to be precise about a concept we had never agreed on. Bias is hard to govern because we lack the principles, not just the tools.

Then the structural claim. As sensors record more behavior and decisions move out of the black box of the human brain into auditable algorithms, **AI systematically expands the terrain that can be subjected to political control**. Sometimes that is good; sometimes bad; but it always raises the stakes of political contest, the way discovering oil inflames a disputed border. We may look back on the pre-AI era as one of enormous autonomy for police officers, managers, teachers, and citizens. And where authorities and citizens are at an uneasy standoff — over surveillance, say — the expansion of control tends to favor the state. Snowden revealed a surveillance regime that had never been publicly debated; Congress then legislated. The literature on digital authoritarianism is the pessimistic reading; a positive agenda for how AI could strengthen liberal institutions is badly under-built.

Great power security competition. The most recalcitrant conflict of all. International relations calls the condition **anarchy**: no authority above states to make or enforce law, so everyone's last resort is force, and everything can be renegotiated or destroyed. Anarchy produces the security dilemma — each state's defense makes others feel less safe — and its bill is enormous: two trillion dollars a year in military spending, thousands of nuclear warheads, a couple of thousand on hair-trigger alert, and plausibly the world's failures on climate, trade, and pandemics too. The obvious remedy, centralization, has its own catastrophic failure mode.

So issues bound up with great-power rivalry are the hardest to govern. The needed institutions are global; consensus at that scale is thin; and even with consensus (everyone agrees nuclear war is bad)

anarchy’s bargaining dynamics can block the institution (the arsenal will not shrink to hundreds, let alone zero). Two clusters live here today. First, military AI: lethal autonomous weapons, cyber operations, influence operations. Norms can hold — the nuclear taboo has since 1945 — but the logic of competition bears down relentlessly; the Pentagon’s long-stated preference for a human in the loop is conceded to be unachievable where instant response is required. Second, decoupling between China and the West: chip supply chains, export controls on lithography, scrutiny of research collaboration, and China’s absence from the main international AI initiatives.

IV. The AI Race

Now the one formula of the lecture.

The metaphor

“AI arms race” is meant to say three things: investment is rising fast, because actors perceive very large geopolitical stakes, and the investment is militarily relevant. The “arms” part is mostly off-point — the real competition is over supply chains, industrial base, science, and prestige, not weapons — and the phrase is routinely abused. Call it strategic technology competition, or simply **the AI race**. What is not in doubt is that great powers believe leadership in AI decides future wealth and power.

The dangerous version is the **race to the bottom**: when actors perceive large gains from relative advantage, they cut corners, exposing everyone to risks they would otherwise avoid. Boeing’s 737 MAX and Uber’s self-driving program are recent commercial examples. The geopolitical nightmare is a crash program — an AI Manhattan Project — in a period of tension, with militaries tempted to deploy powerful but unreliable systems in cyber and kinetic conflict, where speed makes a human in the loop untenable, ML systems misbehave in adversarial settings, and scale means any surprise is a large one.

In math

How risky should we expect such a race to be? The starting model is the **war of attrition**: two players bid up the risk or cost of conflict until one concedes. It is the game of chicken with a continuous action space, and it is equivalent to an **all-pay auction** in which the “revenue” is the risk both players have accepted. With the standard assumptions — rational players, common knowledge of the game — the canonical result is that each player’s expected bid is one third of the value of the prize to that player:

$$\mathbb{E}[\text{bid}_i] = \frac{1}{3} V_i.$$

Read it slowly. V_i is how much player i values winning, measured against the status quo. The bid is measured in the same units. Now suppose a leader perceives the prize as **existential** — as far above the status quo as nuclear war is below it. Then V_i equals the distance from the status quo to nuclear war, and the leader should rationally accept a one-in-three chance of nuclear war to compete. A neighboring model of rent-seeking contests dissipates half the rents rather than a third. So the glass is half full: if the prize is large enough, rational actors will expose the world to a substantial chance of devastation — but they do not race all the way to the bottom.

Conceptual check

Every assumption in that model is false in the real world, and each failure makes the race worse. Take them one at a time.

Common knowledge. Leaders are meeting this game for the first time. Economists joke that the fastest way to raise a hundred dollars is to auction a ten in an all-pay format: someone always fails to realize the right move is not to play, and escalates. The risks are novel tail risks, hard to forecast from experience.

Rationality. Leaders are overconfident and place intrinsic value on winning. When honor or regime survival is perceived to be at stake, backing down costs more than the material stakes.

Observability. The model assumes each side sees the other's risky behavior. If risk-taking has no public signal, or a noisy one, norms of mutual restraint cannot form.

Shared understanding of risk. If AI safety is theory-dependent, each side — through a winner's-curse selection and ordinary rationalization — comes to see its own conduct as safer than the other's, perceives the other as reckless, and escalates. That is a spiral with no rational floor.

If this did not land on first reading: the formula says rational, informed players stop at a third. The check says real players are neither, and stop nowhere in particular.

Escaping the race

The solution space has two ends. **Unilateral** de-escalation: everyone but one concedes, which is what the models assume. Anything that reduces underestimation of risk or makes opting out easier is risk-reducing — though, as usual in coercion models, it may invite the other side to raise its ambition, so the net effect is smaller than the direct one. Force to end a race is also unilateral, and as dangerous as the race.

Cooperative mutual restraint: norms, treaties, and institutions that change the rules of the game so racers internalize the risk. These require agreement on what is unacceptably risky, a way to observe compliance, and incentives (usually sanctions) to enforce it. The core obstacle is the transparency-security tradeoff: enough visibility into the other's arming to reassure, not so much that it compromises their security. Nuclear arms control is the instructive precedent. Third parties — safety organizations, an IAEA or WTO for AI — can articulate focal norms, redirect prestige competition toward pro-social goals as in the Space Race, verify compliance, and resolve disclosure dilemmas; in principle they could even hold hard power, as the Atomic Development Authority was meant to.

But AI is harder to control than nuclear weapons on six counts, and you should be able to recite them: it is more **dual-use**; its non-military **value** is far greater; its assets are more **diffused**; its **risks are harder to discern**; **verification and control** are weaker, since compute is the only tangible chokepoint and it is deeply dual-use; and its **strategic gradient** may stay steep, where nuclear advantage plateaued at secure second strike — the three-hundredth warhead adds little, which is why China stayed under three hundred for decades.

V. Value Erosion

Most of the above is about acute catastrophe. The subtler failure is gradual. Generalize the safety-performance tradeoff: whenever something valuable trades off against performance in a high-stakes contest, competition pushes decision-makers to sacrifice it. Zuckerberg’s prepared response to breaking up Facebook was that it would strengthen Chinese competitors — the logic in one line. Over the long run, competition could proliferate the organizational forms, states, or autonomous systems that lock in impoverished values. The chapter calls this **value erosion**; Bostrom, Christiano, and Hanson each have a version.

The standard objection: history has trended toward human flourishing, so evidently competition does not produce malign evolution. The reply has two parts. First, life did not obviously improve after the Neolithic revolution. Second, post-industrial gains in welfare and liberty may be explained by labor being a **complement** to machinery, so that national power came to depend on an educated, free, supportive citizenry. AI could end that two-hundred-year trend if it substitutes for labor more than it complements it and reduces governments’ need for citizen support. Because erosion is slow it may be easier to observe and coordinate against than a sudden catastrophe; because it is slow, attention may never mobilize in time.

VI. From the Q&A

Is “AI governance” just regulation? No. Regulation is one instrument. Norms inside labs, standards between firms, procurement rules, publication practices, and international institutions are all governance, and most of the leverage today is outside legislatures.

Which lens should I use? All three, in order. Start with GPT for growth and disruption, add information for market structure and power, and use intelligence for alignment and control. The third is where AI stops resembling earlier technologies.

Why does the bias example matter if the classifier can be fixed? Because the fix requires choosing which fairness quantity to prioritize, and that choice is a political one we have not made. The classifier did not create the disagreement; it exposed it.

Does the one-third result mean the race is survivable? It means a rational, informed race is survivable. Every relaxation of those assumptions moves the equilibrium toward the bottom. Treat one third as a floor on risk, not a ceiling.

Why not just build a global institution? Because anarchy is precisely the condition in which you cannot, and because the centralization that would solve anarchy carries its own catastrophic risk. The honest position is that both horns are dangerous and the field’s job is to find the path between them.

Is the “One World or None” framing alarmist? The 1920s analysts who predicted unstoppable gas-bombers were wrong about mechanism and timing, and right about the strategic logic that the nuclear era later confirmed. Confident predictions of nuclear apocalypse were also wrong. The chapter’s closing bet is that they, too, got the logic right: ever more powerful technology and ungoverned great-power competition are not compatible in the long run.

Where to Go From Here

Read the chapter itself; this lecture compressed it. For the GPT lens, Bresnahan and Trajtenberg's original paper is short. For the race, work through the war of attrition in a game theory text until the one-third result is yours. For institutional fit, pick one live domain — frontier-model licensing, compute export controls, content provenance — and run the six-question checklist on the institutions that currently claim it.

The recurring advice is the same as elsewhere in this course: use the frameworks on real cases. It is easy to feel you understand externalities after a lecture and discover otherwise the moment you have to say which institution should internalize which one. Pick a single governance problem and go deep.

Step back, and the lecture reduces to one sentence. AI is a general purpose, information, and intelligence technology — which means it will grow and disrupt, concentrate and empower, and pose problems of alignment and control at every scale — and governing it means building institutions that internalize its externalities under the one condition, anarchy among great powers, in which building institutions is hardest. Tomorrow: what those institutions have actually looked like so far.